MIL-G-62363A(AT)
18 December 1985
Superseding
MIL-G-62363(AT)
15 December 1983

MILITARY SPECIFICATION

GENERATOR, TACHOMETER

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a generator, tachometer for use on a military ground vehicle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards and handbooks. Unless otherwise specified , the following specifications standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6680

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STANDARDS
MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |
| MIL-STD-202 | - Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-462 | - Electromagnetic Interference Characteristics, Measurement of. |
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-45662 | - Calibration System Requirements. |

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

| | |
|---|---|
| 12265503 | - Generator, Tachometer. |

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (6.2), the contractor shall furnish sample generator, tachometers which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representive of the unit to be furnished to the Government. All subsequent generator, tachometers delivered to the Government shall conform to these samples in all their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced standards and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.8.1 and 6.3).



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3.3 Construction. The generator, tachometer shall conform to drawing 12265503. The generator shall consist of three separate components (generator set) a rotor, brush assembly, and stator with keeper (see 4.8.1 and 4.8.2).

3.4 Performance.

3.4.1 Direct current resistance. The direct current resistance of the assembled generator shall be 87.3 ± 12.5 ohms (see 4.8.3).

3.4.2 Insulation resistance. The insulation resistance of the assembled generator shall be greater than 100 megohms at -54 to +25°C (-65 to +77°F) and greater than 25 megohms at +60°C (140°F) with 100 volts (V) direct current (dc) applied between the generator terminals (connected together) and the rotor stack (see 4.8.4).

3.4.2.1 Dielectric withstanding voltage. The electrical insulation shall be capable of withstanding 200 V root mean square (rms), at 60 Hertz (Hz) for 60 seconds between the generator terminals (connected together) and the rotor stack. Leakage current shall be less than 1 milliampere (mA) (see 4.8.4.1).

3.4.3 Sensitivity. Generator sensitivity in either direction in the normal speed range (20.94 milliradian per second to 209.40 milliradian per second) shall be 0.194 ± 0.01 V per radian per second. The tolerance shall include all effects of nonlinearity (see 4.8.5).

3.4.4 Ripple. The generator ripple (average to peak) in either direction or the ratio of maximum voltage deviation to the average voltage expressed as percent ripple voltage shall be ± 7 percent (see 4.8.6).

3.5 Environmental.

3.5.1 Temperature.

3.5.1.1 High temperature (operating). The generator, tachometer shall meet performance requirements 3.4.1 through 3.4.4 during exposure to temperatures up to +70°C (158°F) (see 4.8.7.1.1).

3.5.1.2 Low temperature(operating). The generator, tachometer shall meet performance requirements 3.4.1 through 3.4.4 during exposure to temperatures down to -32°C (-26°F) (see 4.8.7.1.2).

3.5.1.3 Low temperature (non-operating). The generator, tachometer shall not be damaged nor performance impaired after exposure to temperatures down to -54°C (-65°F) (see 4.8.7.1.3).



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3.5.2 Humidity.

3.5.2.1 Operating condition. The generator, tachometer shall meet performance requirements 3.4.1 through 3.4.4 during exposure to relative humidity ranging from 100 percent at +29.4°C (+85°F) to not more than 5 percent at +51.7°C (+125°F) (see 4.8.7.2.1).

3.5.2.2 Non-operating condition. The generator, tachometer shall not be damaged nor performance impaired after exposure to relative humidity ranging from 100 percent at +26.7°C (+80°F) to not more than 2 percent at +68.3°C (+155°F) (see 4.8.7.2.2).

3.5.3 Atmospheric pressure.

3.5.3.1 Operating condition. The generator, tachometer shall meet performance requirements 3.4.1 through 3.4.4 during exposure to an atmospheric pressure equivalent to altitude of not less than 8000 feet (2400 meters) (see 4.8.7.3.1).

3.5.3.2 Non-operating condition. The generator, tachometer shall not be damaged nor performance impaired after exposure to an atmospheric pressure equivalent to altitude of not less than 40 000 feet (12 000 meters) (see 4.8.7.3.2).

3.5.4 Fungus. Materials that are fungus non-nutrients as identified in MIL-STD-454, requirement 4 shall be used in the construction of the generator, tachometer where possible. Fungus growth shall not cause performance degradation (see 4.8.7.4).

3.5.5 Shock (non-operating). The generator, tachometer shall not be damaged nor performance impaired after exposure to 25 gravity units (g) half-sine wave shock impulses for 11 milliseconds (ms) along 3 mutually perdendicular axes (see 4.8.7.5).

3.5.6 Vibration (non-operating). The generator, tachometer shall not be damaged nor performance impaired after exposure to vibration conditions specified (see 4.8.7.6):

a. Simple harmonic motion.
b. Motion applied along three mutually perpendicular directions.
c. Peak acceleration of ± 10 g at all frequencies.
d. Frequency range 20 to 2000 Hz.
e. Frequency sweep varied logarithmically (minimum to maximum only.)
f. Sweep time 35 ± 5 minutes.

Note: In addition the generator shall withstand 5 minutes of vibration at each observed resonant frequency.

3.5.7 Electromagnetic interference. The generator, tachometer shall comply with emissions and susceptability requirements of MIL-STD-461, class A3, Electrical and Electronic Equipment (see 4.8.7.7).



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3.6 Identification and marking. The generator, tachometer shall be marked for identification in accordance with MIL-STD-130 and information provided on Drawing 12265503 (see 4.8.2).

3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspections.

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).
b. Quality conformance inspection (see 4.5).
   1. Examination (see 4.5.2).
   2. Test (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:



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a. Air temperature 23 ± 10°C (73 ± 18°F)
b. Barometric pressure 725 + 50mm mercury (Hg) [28.5 ± 2 inches Hg]
- 75 - 3
c. Relative humidity 50 ± 30 percent

4.3.1 Load resistance. The minimum load resistance for all tests shall be 10 000 ohms with constant speed.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply generator, tachometers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two sample generator, tachometers. Preproduction inspection shall consist of inspection as specified in table I.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two units, from the first 10 generator, tachometers produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table I.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

| Title | Requirement | Inspection | First article | Quality conformance | | Control |
|---|---|---|---|---|---|---|
| | | | | Examination | Test | |
| Material and construction | 3.2 thru 3.3 | 4.8.1 | X | | | X |
| Defects (see 4.5.2 and table II | 3.3, 3.6 and 3.7 | 4.8.2 | X | X | | |
| Direct current resistance | 3.4.1 | 4.8.3 | X | | X | X |
| Insulation resistance | 3.4.2 | 4.8.4 | X | | X | X |



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TABLE I. Classification of inspections.

| Title | Requirement | Inspection | First article | Quality conformance | | Control |
|---|---|---|---|---|---|---|
| | | | | Examination | Test | |
| Dielectric withstanding voltage | 3.4.2.1 | 4.8.4.1 | X | | X | X |
| Sensitivity | 3.4.3 | 4.8.5 | X | | X | X |
| Ripple | 3.4.4 | 4.8.6 | X | | X | X |
| Temperature | 3.5.1 | 4.8.7.1 | | | | |
| High temperature (operating) | 3.5.1.1 | 4.8.7.1.1 | X | | X | |
| Low temperature (operating) | 3.5.1.2 | 4.8.7.1.2 | X | | X | |
| Low temperature (non-operating) | 3.5.1.3 | 4.8.7.1.3 | X | | X | |
| Humidity | 3.5.2 | 4.8.7.2 | | | | |
| Operating condition | 3.5.2.1 | 4.8.7.2.1 | X | | X | X |
| Non-operating condition | 3.5.2.2 | 4.8.7.2.2 | X | | X | X |
| Atmospheric pressure | 3.5.3 | 4.8.7.3 | | | | |
| Operating condition | 3.5.3.1 | 4.8.7.3.1 | X | | X | X |
| Non-operating condition | 3.5.3.2 | 4.8.7.3.2 | X | | X | X |
| Fungus | 3.5.4 | 4.8.7.4 | X | | X | X |
| Shock (non-operating) | 3.5.5 | 4.8.7.5 | X | | | X |
| Vibration (non-operating) | 3.5.6 | 4.8.7.6 | X | | | X |
| Electromagnetic interference | 3.5.7 | 4.8.7.7 | X | | | X |

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the generator, tachometers of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 10 consecutively produced generator, tachometers that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.



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4.5.2 Examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

| Classification | AQL% |
|---|---|
| Major | 1.0 |
| Minor | 2.5 |

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

| Category | Defect | Method of Inspection |
|---|---|---|
| Major | AQL 1.0% Defective | |
| 101 | Dimensions affecting interchangeability, not within tolerance (see 3.3). | SIE 1/ |
| 102 | Improper marking (see 3.6). | Visual |
| 103 | Improper workmanship, affecting inter-changeability (see 3.7). | Visual |
| Minor | AQL 2.5% Defective | |
| 201 | Dimensions not affecting interchangeability, not within tolerence (see 3.3). | SIE |
| 202 | Improper workmanship affecting appearance (see 3.7). | Visual |

1/ SIE = Standard Inspection Equipment.

4.5.3. Tests. Each generator, tachometer assembly shall be subjected to the quality conformance tests specified in table I.

4.6 Control tests. Control tests shall be conducted on one generator, tachometer assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, not less than one test in a six month period. The generator, tachometer assemblies shall be subjected to the control tests specified in table I.

4.7 Failure. Failure of any generator, tachometer assembly to pass any of the specified quality conformance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.



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4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 and 3.3, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.6 and 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Direct current resistance. To determine conformance to 3.4.1, the assembled generator shall have direct current resistance of 87.3 $\pm$ 12.5 ohms under normal room conditions.

4.8.4 Insulation resistance. To determine conformance to 3.4.2, the assembled generator shall have an insulation resistance greater than 100 megohms at -54 to +25°C (-65 to +77°F) and greater than 25 megohms at +60°C (+140°F) with 100 V dc applied between the generator terminals and the rotor stack.

4.8.4.1 Dielectric withstanding voltage. To determine conformance to 3.4.2.1, the dielectric withstanding voltage of the electrical insulation shall be 200 V rms, at 60 Hz applied for 60 seconds between the generator terminals and the rotor stack under normal room conditions.

4.8.5 Sensitivity. To determine conformance to 3.4.3, the generator sensitivity using an output load of 10 000 ohms shall be calculated from the following formula:

$$\text{Sensitivity (volt/radian/second)} = \frac{\text{Volts x seconds}}{\text{revolutions x } 6.28}$$

Tests shall be performed with speeds selected between 6 to 600 milliradians per second. Two tests shall be performed with speeds at least 500 milliradians per second apart.

4.8.6 Ripple. To determine conformance to 3.4.4, the generator percent ripple using an output load of 10 000 ohms, shall be calculated from the following formula:

$$\text{Percent ripple} = \frac{\text{100 x difference voltage}}{\text{average signal level voltage}}$$

Tests shall be performed under normal room conditions at the sample speeds selected in 4.8.5 and two tests conducted as in 4.8.5 with speeds at least 500 milliradians per second apart.



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4.8.7 Environmental.

4.8.7.1 Temperature.

4.8.7.1.1 High temperature (operating). To determine conformance to 3.5.1.1, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 501.2, procedure II. Temperature shall be +70 ± 5°C (+158 ± 9°F).

4.8.7.1.2 Low temperature (operating). To determine conformance to 3.5.1.2, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 502.2, procedure I. Temperature shall be -32 ± 5°C (-26 ± 9°F).

4.8.7.1.3 Low temperature (non-operating). To determine conformance to 3.5.1.3, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 502.2, procedure I. Temperature shall be -54 ± 5°C (-65 ± 9°F).

4.8.7.2 Humidity.

4.8.7.2.1 Operating condition. To determine conformance to 3.5.2.1, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 507.2, procedure II.

4.8.7.2.2 Non-operating condition. To determine conformance to 3.5.2.2, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 507.2, procedure II.

4.8.7.3 Atmospheric pressure.

4.8.7.3.1 Operating condition. To determine conformance to 3.5.3.1, the generator, tachometer shall be tested by placing it in a chamber stabilized at a temperature of +52°C (+126°F) and a pressure equivalent to an altitude of 8000 feet (2400 meters) for not less than 2 hours before the start of the test. After a period of not less than 6 hours in this stabilized condition, and with no change in temperature or pressure, the test article shall meet functional tests.

4.8.7.3.2 Non-operating condition. To determine conformance to 3.5.3.2, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 500.2, procedure I including test for pressure equivalent to 40 000 feet (12 000 meters).

4.8.7.4 Fungus. To determine conformance to 3.5.4, the generator, tachometer shall be tested in accordance with MIL-STD-810, method 508.3. In the event that only materials which are non-nutrients as identified by MIL-STD-454, requirement 4, have been utilized, certification by the manufacturer that only non-nutrients have been utilized will be acceptable in lieu of testing.



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4.8.7.5 Shock (non-operating). To determine conformance to 3.5.5, the generator, tachometer shall be tested in accordance with MIL-STD-202, method 213B, condition J. A peak acceleration of 25g shall apply.

4.8.7.6 Vibration (non-operating). To determine conformance to 3.5.6, the generator, tachometer shall be tested in accordance with MIL-STD-202, method 204D, condition C.

4.8.7.7 Electromagnetic interference. To determine conformance to 3.5.7, the generator, tachometer shall be tested in accordance with MIL-STD-461, class A3, using procedures of MIL-STD-462. During test, the unit shall be operated at all speeds from 20.94 milliradians per second to 209.40 milliradians per second with an output load of 10 000 ohms minimum.

## 5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

## 6. NOTES

6.1 Intended use. The generator, tachometer is intended for use on, but not limited to, the M901 ground-based military vehicle.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Drawing title, number, and date (see 3.3).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If preproduction inspection is required (see 4.4.1).
h. If initial production inspection is not required (see 4.4.2).
i. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Recycled materials. Use of recycled materials which meet the requirements of applicable material specifications without jeopardizing intended use of the item shall be encouraged (see 3.2).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-62363A(AT) | GENERATOR, TACHOMETER FSC6680 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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